Journal of
Marine Science and Engineering

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A Comparison of Methods to Quantify Nano- and/or Microplastic (NMPs) Deposition in Wild-Caught Eastern Oysters (*Crassostrea virginica*) Growing in a Heavily Urbanized, Subtropical Estuary (Galveston Bay, USA)

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Abstract

Nano- and microplastics (NMPs) in waterways reflect the impact of anthropogenic activities. This study examined spatial variations in the presence and types of NMPs in Galveston Bay (Texas, USA) surface waters and eastern oysters (*Crassostrea virginica*). The results reveal most MPs carried by surface waters are fibers > films > fragments. Up to 200 MPs were present in individual oysters [=1.88 (± 0.22 SE) per g wet weight]. Oyster health, based on condition index, varied spatially, but was not correlated with MP load. Based on attenuated total reflectance—Fourier-transform infrared spectroscopy, polyamide and polypropylene were frequently found in waters in the upper bay while ethylene propylene and polyethylene terephthalate were more common in the lower parts of the bay. Pyrolysis–gas chromatography–mass spectrometry revealed a very large range in concentrations of NMPs, from 28 to 10,925 µg ∑NMP/g wet weight (or 172 to 67,783 µg ∑NMP/g dry weight) in oysters. This chemical analysis revealed four main types of plastics present in oysters regardless of location: polypropylene, nylon 66, polyethylene and styrene butadiene rubber. Based on this finding, the average daily intake of NMPs estimated for adult humans is 0.85 ± 0.45 mg NMPs/Kg of body weight/day or a yearly intake of 310 ± 164 mg NMPs/Kg of body weight/year. These findings reveal higher body burdens of plastics in oysters are revealed by the chemical analysis relative to the traditional approach; this is not unexpected given the higher sensitivity and selectivity of mass spectrometry and inclusion of the nanoplastic particle range (i.e., <1 mm) in the sample preparation and analysis.

Keywords: oysters; estuaries; nano- and microplastics; fiber; body burdens

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Academic Editor: Hua Tian

Received: 4 September 2025
Revised: 25 October 2025
Accepted: 27 October 2025
Published: 29 October 2025

Citation: Ciesielski, M.; Hanke, M.; Jurgens, L.J.; Kamalanathan, M.; Mortuza, A.; Gahn, M.B.; Hala, D.; Kaiser, K.; Quigg, A. A Comparison of Methods to Quantify Nano- and/or Microplastic (NMPs) Deposition in Wild-Caught Eastern Oysters (*Crassostrea virginica*) Growing in a Heavily Urbanized, Subtropical Estuary (Galveston Bay, USA). *J. Mar. Sci. Eng.* **2025**, *13*, 2065. https://doi.org/10.3390/jmse13112065

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1. Introduction

Human activities release enormous quantities of plastic debris into waterways. Carpenter and Smith [1] were the first to describe plastics of 2.5 to 5 mm in diameter (polyethylene) floating in surface waters of the ocean (the Sargasso Sea, a region in the North Atlantic). Since then, many studies have focused on the fate and transport of mesoplastics (5–25 mm), macroplastics (>5 mm), microplastics (MPs, <5 mm), and, more recently, nanoplastics (<1 micrometer) found in aquatic ecosystems. The polymers which are found in environmental samples align with the most produced plastics worldwide [2,3]. As part of a meta-analysis (2010–2020), MPs in the environment and food were found to be predominantly composed of polyamide (known as nylon), polyethylene (the most commonly produced plastic), polyethersulfone (a high-temperature engineering thermoplastic), polyethylene terephthalate (a clear, durable, and versatile plastic often used for bottles), polypropylene (which is strong and flexible), poly(vinyl chloride) (also known as PVC which is used extensively in plumbing worldwide) [4], and blue-colored fiber MP particles that are most often found in food. Current research demonstrates that microfibers released from synthetic fabrics washed in laundry machines are particularly problematic in US and global waters near urban centers [5,6]. Fibers and fragments are the major shapes described in environmental studies [3], although filaments, films, pellets, and spheres are also common. Collectively, the annual plastic waste entering the ocean is conservatively thought to be greater than a million metric tons, with many estimates significantly higher than is this estimate [1–6].

We are only now starting to understand the impact of these pollutants of emerging concern on food webs through direct consumption and indirectly, via the plethora of plasticizers they leach, including polychlorinated biphenyls (PCBs), and their adverse consequences to biological systems [5,7–10]. MPs are also considered a source for the potential bioaccumulation of toxins given persistent organic pollutants (POPs) adhere to plastic fragments during the degradation process [11]. In the environment, floating plastic develops surface fouling, which rapidly covers the debris surface with a biofilm, followed by an algal mat, and then colonies of invertebrates in most cases (see [2]). This alters the fate of plastic: it may sink as its density exceeds that of seawater or it may 'swim' as a result of defouling due to organisms foraging on the biofilm and algal mat [2,4–10]. Both processes result in the continued degradation of the plastics, and, alas, its exposure to a broader fraction of the biological community as it sinks or swims through the water column, often repeatedly.

Waterborne MPs enter consumer food webs primarily through suspension-feeding invertebrates, which can then be consumed by estuarine predators and humans. Oysters, which filter enormous quantities of water (3–7 L/h/g dry weight) [12,13], may be a particularly critical linkage between waterborne plastics and an estuary's food webs. However, oysters are selective feeders, with the capacity to actively select food particles with their gills. The extent to which they retain plastic particles in their tissues is an important topic of research. Particle selection processes may allow bivalves to rapidly eliminate MPs from the mantle cavity before being ingested. Nonetheless, MPs and microfibers, typically <100 µm (0.1 mm), have been consistently found in oysters, both wild-caught and farmed, in Europe, Asia, and the U.S. West Coast [5,14,15]. In a recent review of global trends of MPs in oysters, Wotton et al. [16] found that 94% of the oysters examined had MPs, with an average biomass of MPs of 1.41 g wet weight (ww), with wild-caught oysters having more MPs than those grown in aquaculture facilities.

The eastern oyster, *Crassostrea virginica* (Gmelin, 1791), plays a foundational ecological role in estuaries along the Atlantic and Gulf of Mexico coasts (e.g., [17–20]). Historical declines of this species have led to the loss of ecosystem services and decreased commercial

harvests [21,22]. These oyster losses have been attributed to natural environmental factors (floods, hurricane resuspension of sediments, ENSO) [23–26], parasites, viruses (e.g., *Vibrio vulnificus* and *Vibrio parahaemolyticus*), and manmade causes (over-harvesting, dredging of ship channels) in many estuaries [27–30]. In addition, recent studies have shown that marine debris (primarily plastics) accumulation rates are ten times greater in Texas than in other parts of the Gulf of Mexico [31], while earlier studies reported that they were as much as thirty times higher [32,33]. There is also evidence emerging that MPs are entering the largest estuary in Texas (Galveston Bay) [34,35], such that its oysters are at risk.

The goal of this study was to quantify MP loads in oysters and their composition relative to those in the surrounding surface waters. This project used traditional (sorting, counting), modern (attenuated total reflectance–Fourier-transform infrared, ATR-FTIR), and new (pyrolysis–gas chromatography–mass spectrometry with tandem mass spectrometry (Py-GCMS/MS) approaches to examine the spatial variations in the types and numbers of MPs in oysters, and their impact on oyster health. While it was anticipated that these protocols would provide complementary information, given their inherent differences, it was not unexpected that the findings would not be identical. Specifically, traditional approaches rely on counting and categorizing visible particles (and confirming they are plastic), and the ATR-FTIR spectroscopy approach relies on specific particle sizes and shapes that can be measured on the instrument's platform, while Py-GCMS/MS is more likely measuring both nano- and micro-sized particles (NMPs) present in the samples. Py-GCMS/MS provides higher analytical precision in determining the concentrations and types of plastics in oysters. In this study, we aimed to ascertain whether oysters selectively retain certain types of MPs (and NMPs) and examined the potential for MP body burdens in oysters to influence oyster health. While this project mainly characterized variation across space (i.e., regions of the bay), future studies should also consider temporal variations and the sources of MPs.

2. Materials and Methods

2.1. Study Site

Galveston Bay, also referred to as the Trinity-San Jacinto Estuary (29.5° N, 94.8° W), is the drainage basin for the Dallas-Fort Worth metroplex and Houston area watersheds (85,470 km^2). It is composed of four major sub-bays: San Jacinto Bay, Trinity Bay, East Bay, and West Bay (Figure 1). Based on the historical hydrologic record, the Trinity River (55%) delivers the most freshwater inflows, followed by the San Jacinto River (16%) and Buffalo Bayou (12%), with significantly smaller contributions from the remaining tributaries [36]. Exchange with coastal water from the Gulf of Mexico occurs primarily through a narrow pass separating Galveston Island and Bolivar Peninsula. The tidal regime is microtidal (0.15–0.5 m), such that wind forcing and freshwater inflows are important mixing mechanisms in this shallow bay (~2.1 m average).

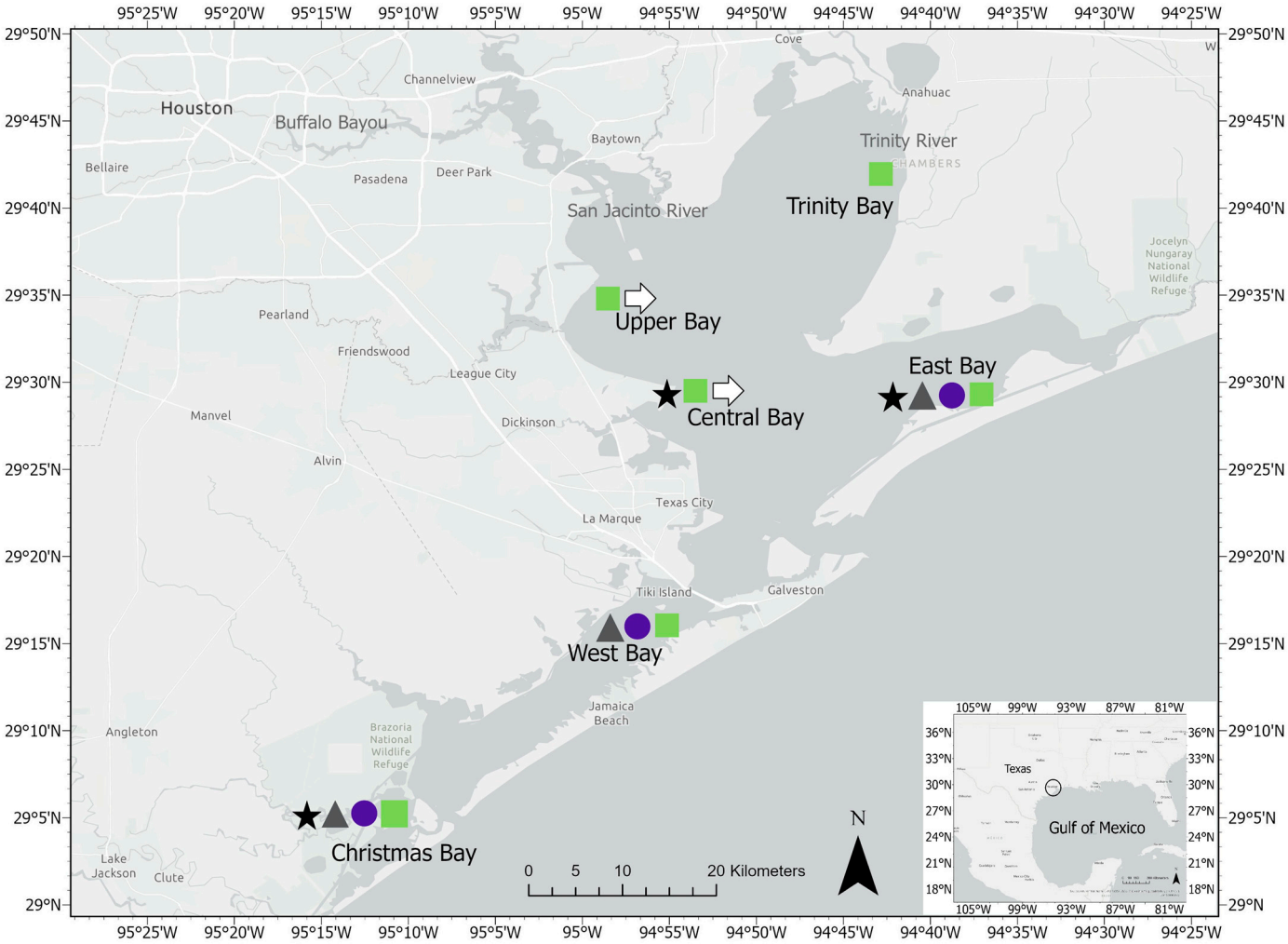

Figure 1. Map of Galveston Bay (Texas, USA) showing the six sampling regions: Trinity Bay (Tbay), Kemah/Seabrook (Ubay), Dickinson/Central Bay (Cbay), East Bay (Ebay), West Bay (Wbay), and Drum Bay, a sub-embayment of Christmas Bay (XBay). Green squares = regions in which surface water was collected, blue circles = regions in which oysters were collected, gray triangles = regions in which oysters were collected for condition index measurements, black stars = regions in which oysters were collected for chemical analysis of MPs in oysters, and arrows = regions in which water samples were collected for chemical analysis. This figure was prepared using ArcGIS Pro 3.5.

2.2. Sampling Design

In Spring 2021, oysters and surface water samples for MP load (counts, type) were collected at three sites, ~100 m apart, in each of six regions: Trinity Bay (Tbay), Kemah/Seabrook (Ubay), Dickinson/Central Bay (Cbay), East Bay (Ebay), West Bay (Wbay), and Drum Bay, a sub-embayment of Christmas Bay (XBay) (Figure 1; Table 1). A GPS was used to georeference each sampling site. Salinity (unitless practical salinity scale), temperature (°C), and dissolved oxygen (mg/L) were measured 0 to 0.5 m below the surface during oyster and water sample collections using a calibrated YSI Pro2030 water quality meter. Wind speed was measured at each site using a Kestrel 2500 weather meter, held vertically for accurate readings and calibrated according to manufacturer recommendations. Oyster samples were checked for vitality (sealed valves); if they were dead, they were left in place. As a result, water samples for MP load were collected from six regions whilst oysters were only collected from three regions (Figure 1; Table 1). When sufficient live oysters were found, 12 market-size (>76 mm) oysters per site were analyzed for MP load. An additional 15 medium-size (50–75 mm) oysters were analyzed for condition index

(CI), which is a proxy for oyster physiological health. In addition, in Spring 2021, surface water samples were collected for ATR-FTIR along a north to south transect in Galveston Bay. For the chemical (Py-GCMS/MS) analysis, ten market-size oysters were collected from three regions in Galveston Bay in Spring 2022: Kemah/Seabrook (Ubay), East Bay (Ebay), and Christmas Bay (XBay).

Table 1. Sample collections across regions of Galveston Bay and number (N) of replicates (total at all three sites).

	Oyster MP Counting	Oyster (Condition Index)	Oyster (Py-GCMS/MS)	Water MP Counting	Water (ATR-FTIR)
Trinity Bay (Tbay)				N = 9	
Kemah/Seabrook (Ubay)				N = 9	N = 4
Dickinson/Central Bay (Cbay)			N = 10	N = 9	N = 2
East Bay (Ebay)	N = 36	N = 20	N = 10	N = 9	
West Bay (Wbay)	N = 36	N = 45		N = 9	
Christmas Bay (Xbay)	N = 34	N = 45	N = 10	N = 9	

2.3. Field Collections—Surface Waters

Three 1 L water samples plus one field blank were collected at each site (Table 1) in DURAN glass bottles (Millipore Sigma, Galveston, TX, USA; Cat no: DWK218010851) that were pre-rinsed with reverse osmosis deionized (RO/DI) water and sealed. Each bottle was lowered below the surface and opened underwater. Bottles were sealed underwater to reduce contamination risks. At each site, a sample blank was also taken. This consisted of opening a pre-rinsed bottle in the air for 15 s. All samples were stored on ice until returning to the lab and then stored at −20 °C until processing. Each field blank was filled with RO/DI water in the laboratory to suspend any collected particles and processed following the same protocols as all water samples. Water samples for ATR-FTIR analysis collected along a transect in Galveston Bay were vacuum filtered (<130 kPa) onto 0.45 µm gridded cellulose nitrate filters (Sigma-Aldrich, Galveston, TX, USA; Cat no: WHA7141104), and the analysis ran as described below.

2.4. Field Collections—Oysters

Oysters were sampled using a small dredge (14.625 m^2: 0.375 m wide dredge) towed from a vessel, or a 0.5 m wide Biloxi-style dredge (Balius Welding and Iron Works, D'Iberville, MS, USA) with 76 mm stretched mesh webbing in the bag, or by hand when water levels were too low for a vessel to operate (Figure 1; Table 1). The dredge was pulled linearly for at least 30 secs at five kmph. Oysters for MP analysis were placed in pre-cleaned polyethylene containers, labeled, and transported to the laboratory on ice, and frozen until processing. 12 market-sized oysters were collected from each of the three sites in West Bay and East Bay (36 total/region), while only 34 were collected from Christmas Bay (Table 1). In addition, while 15 oysters were collected from each of three sites in West Bay and Christmas Bay (45/region) for CI analysis, only 20 oysters were collected from East Bay.

An additional ten oysters were collected from Kemah/Seabrook (Ubay), East Bay (Ebay), and Christmas Bay (XBay) for the chemical (Py-GCMS/MS) analysis of MPs in oysters (Figure 1; Table 1). Oysters were obtained from the Texas Parks and Wildlife Department's (TPWD) annual wildlife monitoring program. Shoreline sampling was conducted using 18.3 m bag seines, while open-water sites were sampled with 6.1 m bay trawls. All collected oysters met or exceeded TPWD's legal size thresholds for recreational and com-

mercial harvest [37]. Specimens were weighed immediately post-capture, chilled on ice, and subsequently frozen at −20 °C at the TPWD Dickinson Marine Laboratory. Samples were later transferred at −20 °C for storage at Texas A&M University at Galveston. Prior to dissection, oysters were thawed on ice. Approximately 1–2 g of mantle and gill tissue were excised from each individual and stored separately at −20 °C until further analysis.

2.5. Condition Index Analysis: Oyster Health Proxy

The condition index (CI) is a commonly used metric to calculate the relative health of an oyster by determining the amount of tissue relative to the available volume inside the shell. Oysters collected for CI analysis were transported to the laboratory on ice and then stored at 4 °C until processing within 24 h. Oysters were pre-cleaned to remove any debris, associated fauna, and any other organisms. Total length and width were measured before determining the wet weight (ww). Oysters were then shucked, and tissues were dried at 70 °C for 24 h, and dried muscle weight measured. Once the shells were completely dried (~2–3 weeks), the shell weight was recorded. CI was calculated using the Hopkins formula [38], with modifications from Abbe and Albright [39] as shown in Equation (1):

$$CI = \frac{dry\ meat\ weight\ (\mathrm{g})}{internal\ shell\ cavity\ volume\ (\mathrm{cm}^3)} * 100 \quad (1)$$

The internal shell cavity volume was determined as the difference in weight between the whole oyster and the weight of the dried empty shell [39,40]. All oysters sampled on each reef measured between 50 and 70 mm shell height. While these oysters were smaller than the market-sized (>76 mm in Texas) ones retained for MP load analysis, oysters outside this size class may have inflated CI values [41].

2.6. Microplastic Analysis—Categorization and Counting

The laboratory methods described herein are based on best practices at the time of the study [42–47]. Water samples were vacuum filtered (<130 kPa) onto 0.45 µm gridded mixed cellulose ester filters (Sigma-Aldrich; Cat no: WHA7141104). If the water sample had high particulate loads, the sample was divided across multiple filters to ensure the filter was not overloaded. Filters were transferred onto Petri dishes, covered, and dried overnight at 60 °C. The dried filters were inspected microscopically, and all MP particles were sorted and counted into categories as detailed in Rochman et al. [5].

Each oyster was rinsed with RO/DI water before the shell was opened to prevent contamination by external MPs. Oysters were then measured with dial calipers along the long shell axis, shucked, and weighed in pre-rinsed weigh boats for wet tissue mass on a precision scale (Mettler Toledo, Columbus, OH, USA; Cat no: 01-804-205). All equipment was rinsed between individual oyster measurements. The oyster tissue was placed into pre-rinsed glass flasks and treated with 150 mL KOH (equivalent to ~10% weight/volume, in a quantity 3 times oyster tissue volume) to extract MPs and incubated for 48 h at 60 °C [45]. After the oyster tissue was digested, the KOH was neutralized with 1 M citric acid solution. The neutralized digestate was immediately vacuum filtered through gridded 0.45 µm mixed cellulose ester filter(s) (Sigma-Aldrich; Cat no: WHA7141104). Ethanol was added (sparingly) to samples with high lipid content to dissolve the fat into solution before filtering using established procedures [48]. Filters were then placed onto Petri dishes, covered, and dried overnight at 60 °C.

Suspected plastic particles were categorized by color and type (e.g., fiber, fragment, film, sphere) using a standardized approach [5]. All MPs on the filter were counted and then a randomized subset of MPs (10%) were imaged using an Olympus DP27 stereo microscope and camera, recording the longest length using ImageJ (version 1.53). To determine which

subset of gridded cells on each filter would be measured, a random number table was used. The lengths of the identified and counted fibers were summed to obtain a total fiber length in millimeters (mm). We assumed the fibers to mainly be nylon 66 (as it was identified to be the most prominent form based on the chemical analysis, see below). Using the average diameter for nylon 66 fibers of 17 µm (or 0.0017 cm) [49], we calculated a radius of 0.00085 cm (i.e., 0.0017/2 = 0.00085 cm). Using the length of the fiber (which was converted to cm by dividing the length in mm by 10) and its radius, we calculated their volume in cm^3 assuming fibers are cylinders: $\pi*r^2*h$ (where π = pi, r^2 is the radius squared, and h is the height or total length of the fiber). Next, the mass of fiber (in grams) was calculated by multiplying the specific density of the nylon 66 fiber (1.14 g/cm^3; [50]) by the previously calculated volume in cm^3. Finally, the fiber mass was expressed as a concentration in µg/g ww oyster gill/mantle tissue by accounting for 8% of the total oyster mass comprising the gill/mantle tissue (personal observations; Aurora Gaona-Hernández). The concentration (in ww) was converted to a dry weight (dw) using a conversion factor established in Mortuza et al. [10]; this protocol involved overnight lyophilizing gill/mantle tissue prior to analysis. The conversion of nylon 66 length data to a concentration in µg ww/g ww (or dw) enabled a qualitative comparison of the nylon fiber body burden with that of nylon 66 NMP tissue bioaccumulation as quantified using Py-GCMS/MS.

2.7. Microplastic Analysis—Chemical Analysis

A subsample of MPs collected from surface waters, spanning a variety of colors and sizes, was identified using attenuated total reflectance–Fourier-transform infrared (ATR-FTIR) spectroscopy [51]. Spectra were obtained on samples using an Agilent Technologies Cary 630 FTIR controlled by Agilent MicroLab FTIR Software Version 1.0.07. Absorbance spectra were collected from 3650 to 600 cm^{-1} at a spectral resolution of 8 cm^{-1} with 50 scans co-added as described in detail in Kamalanathan et al. [52]. During analysis, care was taken to ensure that the plastic samples covered the diamond crystal on the sampling module, and the pressure arm was rotated downward to apply sufficient contact pressure. The FT-IR spectra were then uploaded to https://www.openanalysis.org/openspecy/ (last accessed 2 April 2025) for microplastic characterization. This applies a baseline correction, smoothing, normalization, and wavenumber alignment and runs correlation-based matching against downloaded libraries.

In addition, frozen oyster tissue (~0.5 g) collected for chemical analysis using an Agilent pyrolysis–gas chromatography–mass spectrometry with tandem mass (Py-GCMS/MS) spectrometry [10]. Tissues were lyophilized (24 h, LABCONCO freeze dryer, Kansas City, MO, USA) and then pulverized before an enzyme solution (1 mL; 0.06 g Pez/mL buffer, pH 8) was used to digest them for 24 h at 40 °C [53]. They were then filtered through 2.8 µm and 0.7 µm furnaced Whatman (Cat no: WHA1825047) glass fiber filters (GFFs), rinsed (Milli-Q water), and dried (24 h, 38 °C). The GFFs were then placed in separate stainless-steel pyrolysis cups (Eco-Cup LF, Frontier Labs, Austin, TX, USA) pretreated with 20 µg of polyfluorinated styrene as internal standard and topped with 5 mg of $CaCO_3$ and quartz wool. A Frontier Laboratories Auto-shot sampler pyrolyzer in conjunction with an Agilent 8890 GC System coupled with an Agilent 7010B triple quadrupole mass spectrometer was used to identify and quantify these nano- and microplastic (NMP) polymers: polymethyl methacrylate, polypropylene, polyvinyl chloride, polyamide, polycarbonate, nylon 66, polyethylene, polyethylene terephthalate, acrylonitrile butadiene styrene, polyurethane, styrene-butadiene rubber, and polystyrene [10]. Pyrolyzates were separated using an Ultra Alloy$^+$-5 Capillary Column in an Agilent 8890 GC System with an increasing temperature gradient before being introduced to an Agilent 7010B GCMS/MS Triple Quadrupole (Austin, TX, USA) for mass analysis [10]. Each polymer was quantified using a linear

five-point calibration curve containing an internal standard (polyfluorinated styrene) [10]. Possible interference by lipids on plastics quantification was corrected as described by Mortuza et al. [10]. Blank controls, recovery controls, and replicates were also analyzed. The % recovery of notable plastics in the muscle matrix was 115 ± 0.15% for polypropylene, 93 ± 0.08% for nylon 66, and 113 ± 0.05% for polyethylene [10].

Seafood daily intake (DI) and MP average daily intake (ADI) rates were calculated as follows:

$$DI = \frac{DC_{shellfish\ or\ fish}}{BW} \tag{2}$$

$$ADI = C_{Plastics} * DI \tag{3}$$

The seafood daily intake (DI; g/Kg human body weight/day) was determined by dividing the estimated daily consumption ($DC_{shellfish/fish}$ of shellfish; 30 g/day) by the body weight (BW) of an average human adult (70 kg; Equation (2)) [54]. This was used to determine an average daily intake (ADI) rate of plastics (mg/Kg human body weight/day) by multiplying the average plastic concentration in oysters (µg/g ww; $C_{plastics}$) by the DI (Equation (3)).

2.8. QAQC, Controls and Blanks

Contamination of samples by extraneous MPs is a common issue in both field and laboratory studies. Quality assurance and control measures were taken to reduce and monitor the degree of MP contamination in all samples collected (see the review by Shumway et al. [8]). Maroon-colored lab coats were worn while processing samples. Because of this, any maroon fibers were removed prior to data analysis. In addition, lab surfaces were cleaned daily with non-shedding cloths before any sample processing occurred. All efforts were taken to limit mixing of the water prior to collection, including drifting to the site, and opening sample bottles under water. Field blanks were collected concurrently to samples; MPs present in these blanks were quantified and used to correct the water MP counts. Only live, closed oysters were retained for MP analysis, ensuring that internal tissues were not exposed during field collections. Prior to shucking, oyster shells were cleaned with RO/DI water to prevent contamination by MPs on the shells. To further prevent contamination, glassware and tools (e.g., forceps) were rinsed three times with RO/DI water and samples were kept covered with aluminum foil during digestion. A procedural blank was included in each round of sample processing; this blank underwent digestion and filtration steps. After filtration, filters were placed in clean Petri dishes and covered. The filters remained covered through the remainder of processing and only opened if additional verification was required. If present, any MPs found on the procedural blanks were recorded. In doing so, the goal was to limit bias by treating the procedural blanks identically to the samples. These blank values were used to correct sample values for contamination by MPs. As part of the categorization and counting process, procedural blanks (2), consisting of only 10% KOH, were prepared in tandem with each digestion. MP counts were adjusted using these blanks, which were calculated by subtracting the mean number of plastics in procedural controls for each color-type category from the oyster totals for that round before analysis. If particles had been misidentified, counts were corrected. To ensure the precision of MP counts, a subset of samples was counted by two individuals. These counts were compared and included herein only if they were within 10% of each other. If not, the sample was recounted by both individuals.

2.9. Data Analysis

Statistical analyses were performed in R version 4.1.2 with the lme4 package [55]. At each site, the mean and standard error (SE) were calculated to determine variability across

sites and regions. We applied a mixed-effects modeling approach to analyze MP patterns by site, region, and sample type, including procedural controls. We used a negative binomial distribution and incorporated site (within region) and individual oyster mass as random intercepts (tissue mass range: 5–78 g). The ggplot boxplot in R displays the median, and 25th and 75th percentiles. Data beyond the end of the whiskers are outliers and plotted individually. In SAS (Version 9.4), an analysis of variance (ANOVA) was used to determine if there were statistically significant differences in oyster CI based upon the reef the oysters were collected from. Before running this GLM procedure with random effects to account for the variation in total number of oysters collected among the individual reefs, homogeneity of variance was tested utilizing Levene's test. The data met normality assumptions, and no transformations were necessary. A Student–Newman–Keuls (SNK) post hoc test was applied to determine how reefs varied from one another. A Pearson's product-moment correlation analysis was used to test for significant ($p < 0.05$) relationships between mean oyster condition and mean plastic count from the water filter samples and between the mean oyster condition and the mean plastic count from the oyster filter samples.

3. Results

3.1. Water Quality

Samples were collected at three sites in each of the six regions across Galveston Bay (Figure 1). Salinity increased from north to south, from Trinity Bay (13.57 ± 0.93) and Kemah/Seabrook (Upper Bay) (16.30 ± 1.0) in the upper reaches of the bay, to Dickinson/Central Bay in the middle (16.73 ± 0.47). The highest salinities recorded in Galveston Bay were in the lower reaches of the bay, which are under the influence of tidal exchange with the Gulf of Mexico, that is, East Bay (20.97 ± 1.13), West Bay (21.77 ± 0.17), and Drum Bay, a sub-embayment of Christmas Bay (24.43 ± 0.07) (Table 2). Temperature (17.5–24.0 °C) and dissolved oxygen (6.48–8.81 mg/L) were measured just below the surface; these were similar across all collection sites and regions (Table 2). Wind speed was highly variable, from 2.43 m/s (± 0.57) to 4.90 m/s (± 0.67) (Table 2). Despite searching for oysters in all six regions of the bay in 2021 where they have been known to occur historically, live oysters were only found in three regions in the lower reaches of the bay: East Bay, West Bay and Christmas Bay (Figure 1; Table 1). The oyster CI varied significantly among the reefs sampled ($F_{7,101} = 19.47$, $p = <0.001$), with significantly higher oyster CI in East Bay compared to West Bay and Christmas Bay (Figure 2).

Table 2. Water quality conditions measured during sample collections (Spring 2021) of the six regions throughout Galveston Bay. Salinity is reported on the unitless practical salinity scale. Average values (±SE) were calculated for nine measurements in each region, each which had three unique sampling sites.

	Salinity	**Water Temperature (°C)**	**Dissolved Oxygen (mg/L)**	**Wind Speed (m/s)**	**Depth (m)**
Trinity Bay (Tbay)	13.57 ± 0.93	22.40 ± 0.20	7.25 ± 0.006	3.97 ± 0.32	2.90 ± 0.15
Kemah/Seabrook (Ubay)	16.30 ± 1.0	19.90 ± 0.10	8.32 ± 0.08	2.97 ± 0.23	3.60 ± 0.06
Dickinson/Central Bay (Cbay)	16.73 ± 0.47	18.17 ± 0.24	8.81 ± 0.267	2.43 ± 0.57	2.76 ± 0.63
East Bay (Ebay)	20.97 ± 1.13	19.87 ± 0.07	7.96 ± 0.05	4.13 ± 0.33	1.78 ± 0.22
West Bay (Wbay)	21.77 ± 0.17	17.50 ± 0.20	8.58 ± 0.02	4.87 ± 0.12	1.50 ± 0.24
Christmas Bay (Xbay)	24.43 ± 0.07	24.00 ± 0.17	6.48 ± 0.13	4.90 ± 0.67	0 ± 0

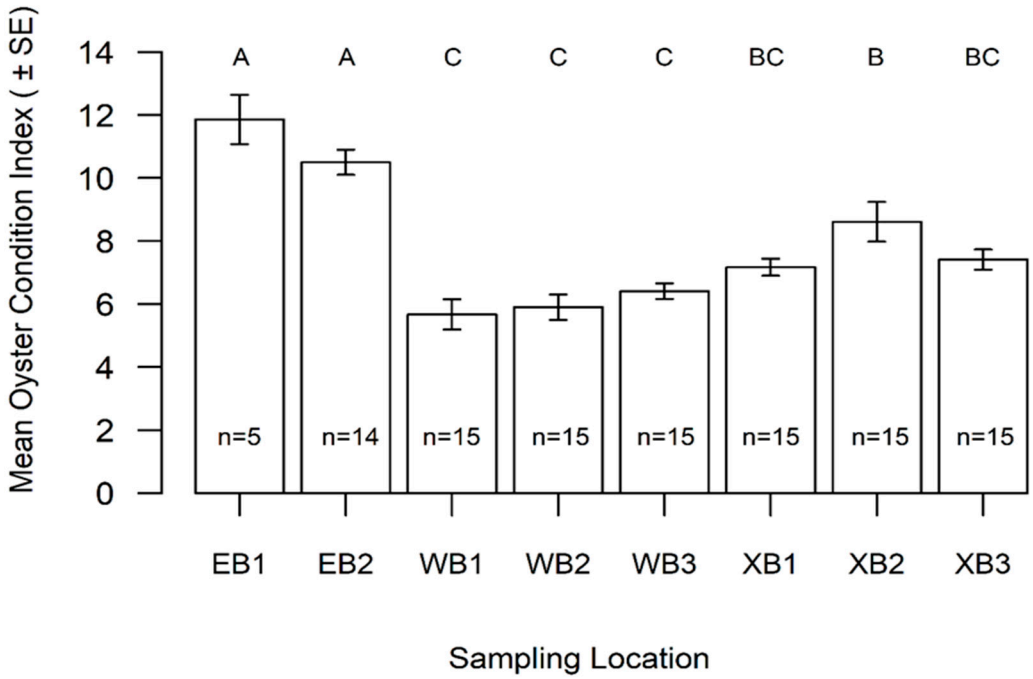

Figure 2. Oyster Condition Index values (mean ± SE) for individual reefs sampled from East Bay (EB), West Bay (WB), and Christmas Bay (XB). The total number of oysters measured in each location (n) is also provided. Reef three in East Bay was not included in the analysis because there were insufficient live oysters present. Letters show significantly different results ($p < 0.05$) based on an SNK post hoc test.

3.2. Microplastics—Field, Sampling and Procedural Blanks

All field collection and procedural blanks contained relatively low numbers of MPs (Figure 3). Field collections (water blanks) from each site/region were similar; there was no distinct north–south gradient observed in the number of MPs, except for Christmas Bay which had the lowest median number of MPs in the blanks (Figure 3a). Field collection blanks from the Trinity River delta and West Bay had the greatest range (min–max = 22 to 84 and 21 to 85, respectively) with a median of 28 MPs per L and 44 MPs per L, respectively. In the other regions of the bay, the median number of MPs varied from 12 to 52 MPs per L (Figure 3a). Oyster procedural blanks had similar median values (28) (Figure 3b). Given the variability (see outliers) in blank contamination for each processing round for oysters, MP counts were adjusted based on the blanks associated with the specific processing round to minimize propagating errors. Overall, the field collection blanks had a median of 20 MPs per L (min–max = 6–54); the vast majority of these were clear fibers at each location, most likely from the RO DI filter itself. The RO DI water blank had a median of 13 MPs. The fiber dominance could also have indicated air pollution; however, additional field and procedural blanks revealed this was not the case. The RO DO data can be found in the data repository (https://doi.org/10.18738/T8/ML0CHU).

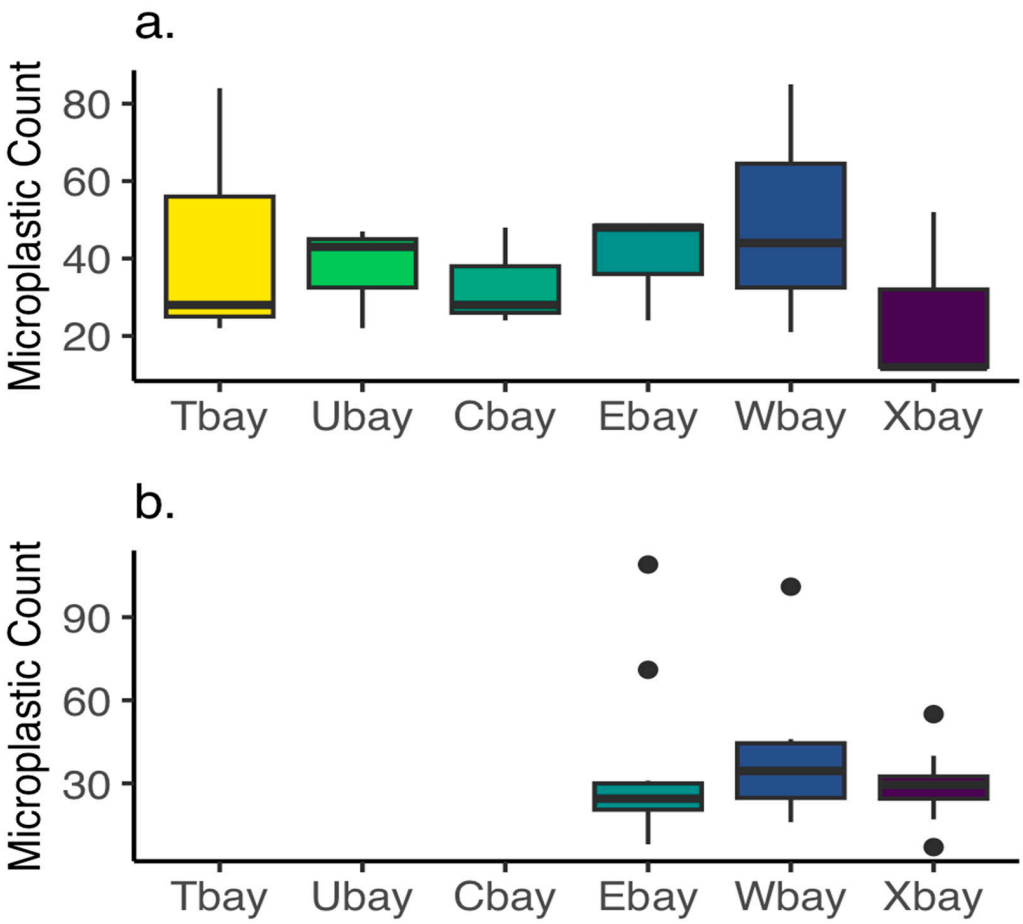

Figure 3. Microplastic counts in the (**a**) field collection blanks (water sampling) and (**b**) procedural blanks (oyster processing). Samples were collected from Trinity Bay (Tbay), Kemah/Seabrook (Ubay), Dickinson/Central Bay (Cbay), East Bay (Ebay), West Bay (Wbay), and Drum Bay, a sub-embayment of Christmas Bay (XBay). Box and whisker plots include median values (bar), 25th, and 75th percentiles, with outliers plotted individually.

3.3. Microplastics—Surface Waters

The total counts of MPs in surface water samples collected from across the bay were lowest in Christmas Bay (median = 11 MP/L) and in Central Bay (median = 18 MP/L) (Figure 4a). In all other regions, median MP counts were similar: Trinity Bay (29 MP/L median, min–max = 8–78), Upper Bay (37 MP/L median, min–max = 12–81), East Bay (22 MP/L median, min–max = 9–57), and West Bay (24 MP/L median, min–max = 4–57). While there were regional variations, the distribution of MPs did not appear to follow a gradient in the bay.

Of the MPs in surface waters, microfibers were found to be the most prevalent form throughout the bay (Table 3, Figure 4b), with fragments and films making up the majority of non-fiber MPs counted (Table 3, Figure 4c). There appeared to be more fibers in the upper parts of the bay—Trinity Bay and Upper Bay—relative to the lower bay—Central Bay, East Bay, West Bay, and Christmas Bay—but these findings were not significant (Figure 4b). Clear microfibers (609) were the most common color present, in much greater abundance than > black (254) > blue (149) > gray (38) > pink, green, orange, purple, yellow, brown (Table 3). When present, films were found throughout the bay (Figure 4c). Pink, red, and blue fragments were more common than those of other colors (Table 3; Figure 4c). Together, these accounted for more than 95% of the fragments counted.

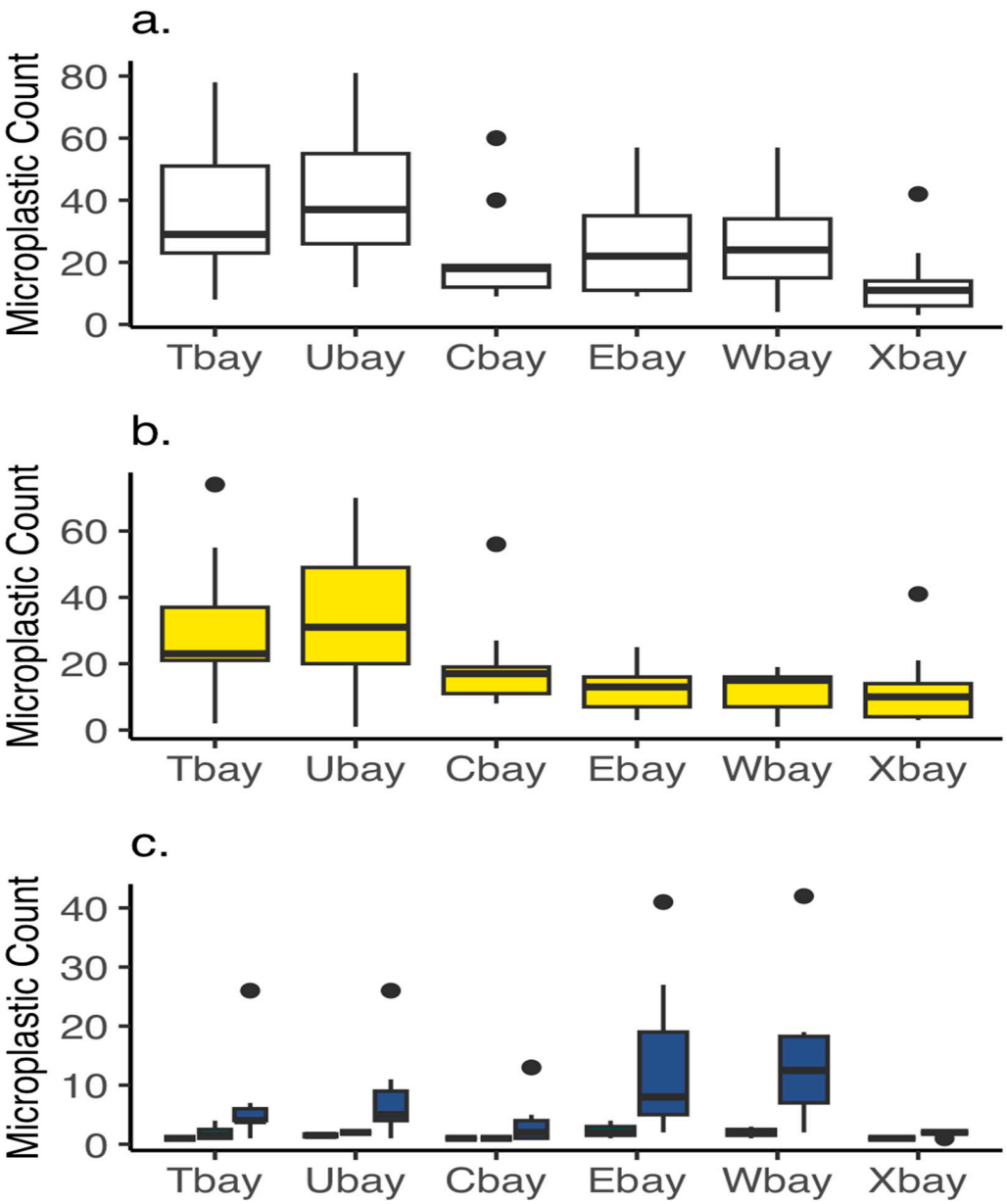

Figure 4. Microplastics in surface water samples (**a**) across all bay regions, and the dominant types of MPs which includes (**b**) fibers (yellow), and (**c**) fragments (blue), films (teal) and bundles (green). Samples were collected from Trinity Bay (Tbay), Kemah/Seabrook (Ubay), Dickinson/Central Bay (Cbay), East Bay (Ebay), West Bay (Wbay), and Drum Bay, a sub-embayment of Christmas Bay (XBay). Box and whisker plots include median values (bar), 25th, and 75th percentiles, with outliers plotted individually.

Table 3. Counts of MPs present in surface waters and oysters. MPs were sorted by color and type. While fibers, film, and fragments were the most common types, occasionally other forms of MPs were also present. Gray cells reflect MPs of a particular category that were absent. The other category includes bundles and spheres.

	Water				Oyster Tissues			
Color	**Fiber**	**Film**	**Fragment**	**Other**	**Fiber**	**Film**	**Fragment**	**Other**
Black	254	2	12		347	8	22	1
Blue	149	5	75	1	219	8	45	
Brown	2				5			1
Clear	609	11	23	3	2415	57	148	31

Table 3. *Cont.*

	Water				Oyster Tissues			
Color	**Fiber**	**Film**	**Fragment**	**Other**	**Fiber**	**Film**	**Fragment**	**Other**
Green	4		3		12	2	3	
Gray	38		5		64	2	2	
Multi			5					
Orange	5		1		10			
Pink	12		197		28	4	9	
Purple					7		8	
Red		6	86			11	118	
Silver					1			
White	1		1	1	4	2	4	1
Yellow	7				10		1	

3.4. Microplastics—Oysters

After subtracting the numbers of MPs in the associated procedural blanks, the total number of MPs per oyster ranged from 0 to 200 (Figure 5a,b), with a median of 23 MPs/oyster and an average of 34.1 (±3.6) MPs/oyster. There was no significant effect of region on MP abundance (Figure 5a,b). However, site-level variation within regions was significant ($p < 0.0001$). Site-level differences were largely driven by two sites in Christmas Bay and West Bay ($p < 0.0001$, $p = 0.0097$, respectively). In oysters, the following MPs were commonly observed: microfibers, fragments, and films (Table 3; Figure 5). There were also bundles and spheres but because these were a minor fraction of the total counts, we summed them as "other". Microfibers were the most abundant form of MPs encountered in Galveston Bay, with clear microfibers present at several orders of magnitude higher than both other microfiber colors and other MPs. Clear (2415) > black (347) > blue (219) > gray (64) > pink, green, orange, purple, yellow, brown, while silver microfibers were the most frequently observed fibers (Table 3; Figure 5a,c). The majority of MP fibers in oysters were between 500 and 2000 µm in length (see https://doi.org/10.18738/T8/ML0CHU for data). Fragments, defined here as irregularly shaped, hard and inflexible, rigid edges, and not breaking when compressed, were the second most common MPs present in oysters in the bay (Table 3; Figure 5b,d). Of these, clear (148) and red (118) fragments were the most prevalent forms, accounting for >80% of fragments (Table 3). Blue, black, pink, purple, white, green, yellow, and gray fragments were also commonly found. Clear films were the most common color, accounting for >85% of all films. While other MPs were observed, they were present infrequently (Table 3). The mean number of MPs per gram of oyster ww across Galveston Bay was 1.88 (±0.22) g ww (Figure 5c,d). MP counts did not appear to depend on oyster mass (AIC: 957.5 vs. 944.9). There was no significant correlation between the average oyster CI and the mean number of plastic counts per filter for the same reef ($r_6 = -0.32$, $p = 0.42$), total plastic count per liter of water collected over the reef ($r_6 = -0.18$, $p = 0.65$), and average count of plastic pieces found within oyster tissue ($r_6 = -0.18$, $p = 0.65$).

Finally, the conversion of fiber lengths to a body burden yielded an average concentration of 9.10 ± 1.17 µg/g ww (min–max, 0.13–55.36 µg/g ww) or 56.43 ± 7.28 µg/g dw (min–max, 0.78–343.43 µg/g dw). While not directly comparable to the chemical analysis performed using Py-GCMS/MS (as described below), the conversion of count data provides a qualitative comparison between the two methods, while keeping in mind the expectation that the count data is likely to underestimate the total bioaccumulation of NMPs in oysters (due to its reliance on visual analysis and manual sorting).

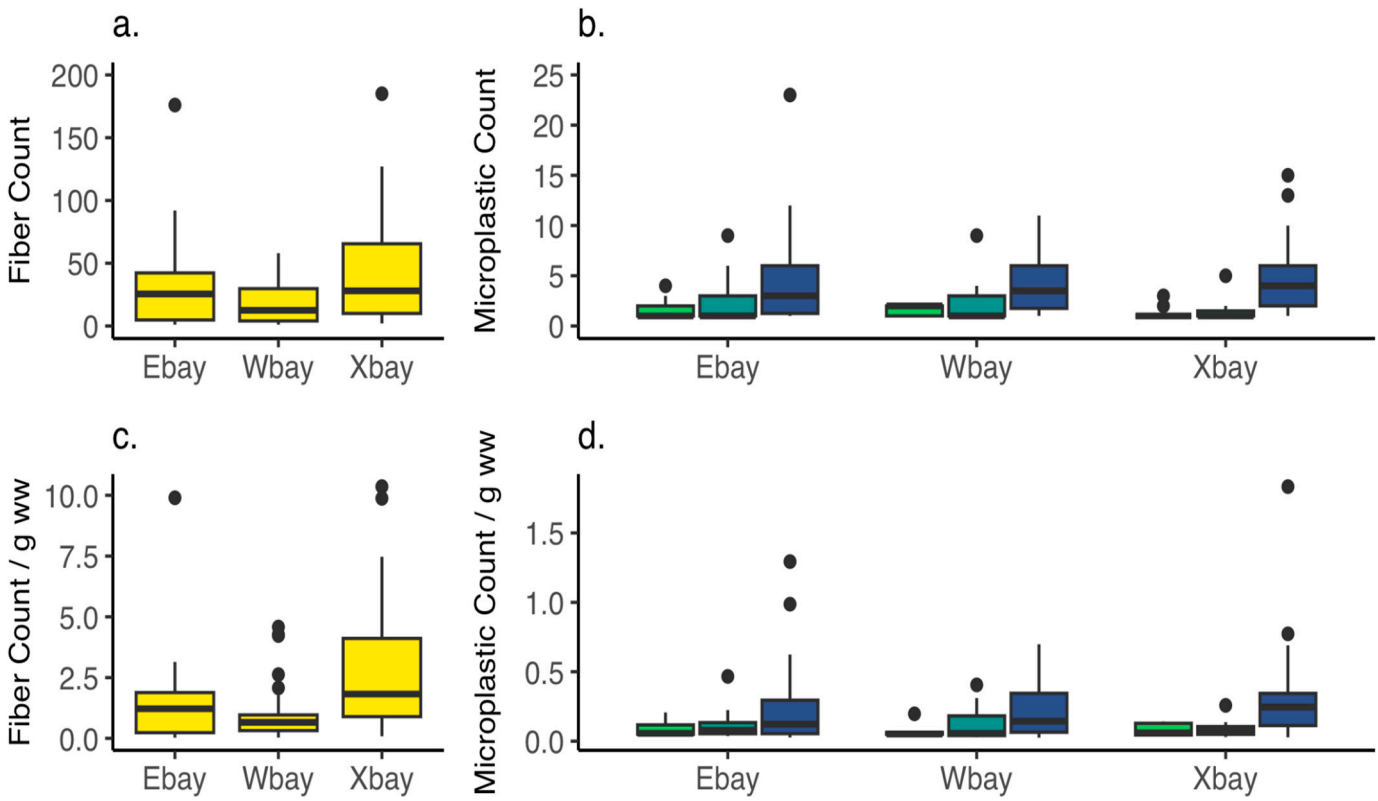

Figure 5. MP counts per oyster (n = 12) by site after subtracting the relevant procedural blank for (**a**) fibers (yellow), and (**b**) films (teal), fragments and spheres (blue) and bundles (green). MP counts per g of oyster of tissue (wet weight) calculated for (**c**) fibers (yellow), and (**d**) films (teal), fragments and spheres (blue) and bundles (green). Oysters were sampled from East Bay (Ebay), West Bay (Wbay), and Christmas Bay (Xbay). Box and whisker plots include median values (bar), 25th, and 75th percentiles, with outliers plotted individually.

3.5. Chemical Analysis of Microplastics—ATR-FTIR

ATR-FTIR spectra of MPs in the water were compared to the reference database developed by Primpke et al. [56]. As seen in Figure 6a, FTIR spectra showed peaks at 1650, 1518, and 1458 cm^{-1} corresponding to bands amide I and CH_2 bend of polyethylene terephthalate (PET) with a correlation of r = 0.53 (Table 4). Peaks at 2917, 2850, 1460 and 1375 cm^{-1} corresponding to ν_{as} CH_2, ν_s CH_2, and CH_2 bend of ethylene propylene with a strong correlation of r = 0.86 and peaks for 2917, 2850, and 1460 cm^{-1} ν_{as} CH_2, ν_s CH_2, and CH_2 bend of polyethylene with a correlation of r = 0.55 (Figure 6a,b; Table 4). In another sample, peaks at 2951, 2917, 2867, 2837, 1450, and 1375 cm^{-1} corresponding to ν_{as} CH_2, ν_s CH_2, CH_2 bend and amide III of polypropylene with a strong correlation of r = 0.71 and peaks at 1635 and 1540 cm^{-1} corresponding to amide I of polyamide with a correlation of r = 0.55 (Figure 6a,b; Table 4). Based on ATR-FTIR, polyamide and polypropylene were frequently found types of MPs in the upper parts of the bay while ethylene propylene and polyethylene terephthalate were more commonly found in the central parts of the bay (Table 4).

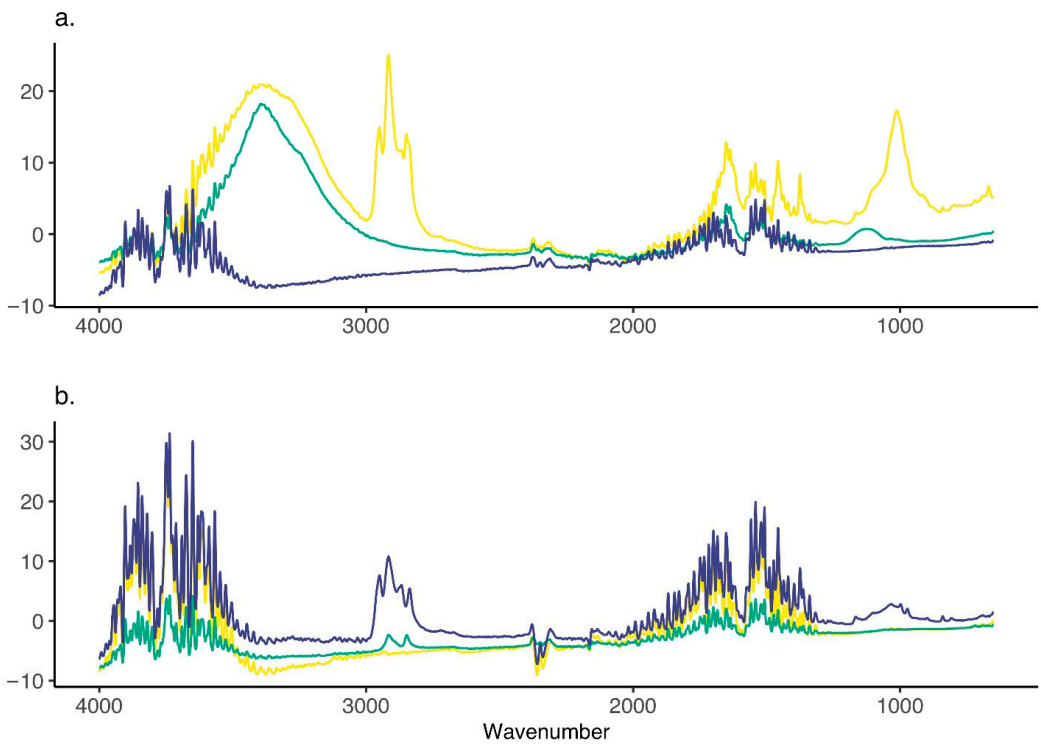

Figure 6. ATR-FTIR spectra of representative MPs examined from surface waters in Galveston Bay: (**a**) Ethylene propylene (yellow), polyethylene terephthalate (PET; green), and polyamide (PA; blue), and (**b**) polypropylene (PP; blue), polyethylene (PE; green), and silicone (yellow). Samples and locations are given in Table 1; sample identification is given in Table 4.

Table 4. Summary of representative MPs identified using ATR-FTIR spectra of surface water samples collected in Galveston Bay as part of a preliminary study in Spring 2020. Locations are shown in Figure 1. Reference spectra in Primpke et al. [56] were used to confirm the identity of the MPs. Pearson's r provides the confidence level of a correct identification.

Sampling Site	Location	Material Identified	Pearson's r
Station 1	Upper Houston Ship Channel (Ubay)	Polyamide	0.55
Station 1	Upper Houston Ship Channel (Ubay)	Polypropylene	0.71
Station 2	Morgan's Point (Ubay)	Polyamide	0.54
Station 3	El Jardin (Ubay)	Silicone	0.61
Station 4	Seabrook (Ubay)	Polyethylene	0.55
Station 4	Seabrook (Ubay)	Ethylene propylene	0.86
Station 5	San Leon (Cbay)	Polyethylene terephthalate	0.53
Station 6	Texas City (Cbay)	Silica gel	0.56

3.6. Chemical Analysis of Microplastics—Py-GCMS/MS

NMPs found in oysters were analyzed using Py-GCMS/MS, which revealed a very large range in concentrations from 28 to 10,925 µg ∑NMP/g ww oyster biomass (or 172 to 67,783 µg ∑NMP/g dw oyster biomass), with the highest median concentrations in Christmas Bay compared to Upper Bay and East Bay (Figure 7a). This pattern in NMPs in oysters is similar to that observed when counting individual MPs (Figure 5). The chemical analysis revealed four main types of NMPs present in Galveston Bay oysters regardless of location. These were, from highest to lowest concentration, polypropylene (PP) (59% of total NMPs), nylon 66 (N66) (27% of total NMPs), polyethylene (PE) (12% of total NMPs),

and styrene butadiene rubber (2% of total NMPs) (Figure 7b). Of these, we observed similar distributions of polyethylene and nylon 66 in oysters regardless of location, but higher concentrations of polypropylene in Christmas Bay relative to the other two regions and higher concentrations of styrene butadiene (rubber) in East Bay than in Christmas Bay (Figure 7b). While six other common plastic materials were measured, they were not found in detectable concentrations in Galveston Bay oysters. Based on these findings, we calculated the average daily intake (ADI) of NMPs for adult humans to be on average 0.85 ± 0.45 mg NMPs/Kg of body weight/day, and the yearly intake was estimated to be an average 309.98 ± mg NMPs/Kg body weight/year for MPs. The minimum to maximum range of ADI's was 0.01 to 4.80 mg NMPs/Kg of body weight/day, and yearly intake was 4.45 to 1752.60 mg NMPs/Kg body weight/year for MPs, respectively.

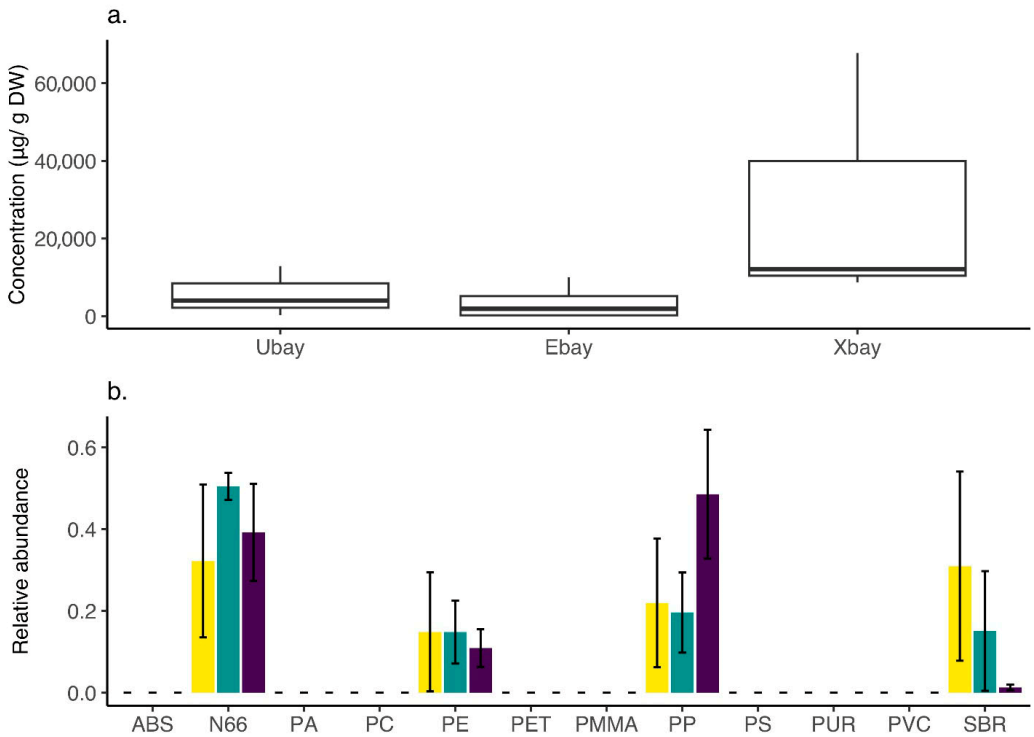

Figure 7. MPs in oysters were measured with pyrolysis–gas chromatography–tandem mass spectrometry (Py-GCMS/MS). (**a**) Concentrations (µg/g dry weight) in oysters collected from Kemah/Seabrook (Ubay), East Bay (Ebay), and Drum Bay, a sub-embayment of Christmas Bay (XBay). (**b**) Relative abundance of polymethyl methacrylate (PMMA), polypropylene (PP), polyvinyl Chloride (PVC), polyamide (PA), polycarbonate (PC), nylon 66 (N66), polyethylene (PE), polyethylene terephthalate (PET), acrylonitrile butadiene styrene (ABS), polyurethane (PUR), styrene-butadiene rubber (SBR), and polystyrene (PS) in oysters from Ubay (teal), Ebay (yellow), and Xbay (purple). All values are mean ± standard error. Concentrations are normalized to total to show relative abundance.

Finally, the comparison of the average fiber count data converted to body-burden concentrations (i.e., 9.10 µg/g ww or 56.43 µg/g dw) with the Py-GCMS/MS data for nylon 66 (i.e., 5177 µg/g ww or 32,087 µg/g dw) showed the analytical chemical values to only be 57x higher than those estimated from the count data. This is a relatively close approximation, that is, within two orders of magnitude, given the disparity of analysis between the two methods (i.e., observational counts vs. analytical chemical determination). The relative agreement between the two methods is likely a reflection of the overall similarity in the tissue processing step involving the filtration of digested oyster tissue through a 0.45 µm cellulose filter (as used in the categorization and counting method) or 2.8 µm and

0.7 µm GFFs (as used in the chemical analysis method). We can expect this step to be key towards retaining NMPs for downstream analysis.

4. Discussion

As anthropogenic stressors continue to impact ecological components of estuarine communities, MP consumption by commercially important bivalve species may impact their health, as well as potentially human health [6,7,16,57,58]. Given the Gulf of Mexico has been reported to have some of the world's highest MP densities [59], comparable to the Great Pacific Garbage Patch in some places, understanding their impacts on biological communities in this region is critical. Further, the recent study by Wessell et al. [31] indicated that Texas was ten times more likely to be influenced by marine debris than other estuaries in the Gulf. This study combines three different measurements of MPs. In the first part of the discussion, we focus on the count-based method, while in the second part, we focus on the chemical analysis, and with it the measurement of concentrations (both nano and micro). In both cases, these procedures gave findings which are consistent with other published observations.

For example, we found MPs were abundant in the water and oysters sampled in our study, with a mean of 28.1 (±2.67) MP/L in water and 1.88 (±0.22) MPs/g ww tissue in oysters, respectively, based on the traditional approach. The overall density of MPs in water samples in Galveston Bay reported herein is within the range reported in an earlier study conducted by Oakley et al. [35] in this estuary, which found an overall density of 44.5 MPs/L, with a range of 16.8 to 72.2 MPs/L. Oakley et al. [35] reported fewer MPs in water grab samples if sampling was conducted after a period of low to no freshwater inflows compared to a period of high flows. This suggests riverine sources of MPs to this estuary, which is consistent with predictions in Summers et al. [34]. This will be discussed more below. Our findings are also within the ranges previously reported in oysters elsewhere in the US (up to 140 MPs/oyster or 0.18 to 18.58 MPs/g ww tissue [60–63]). These earlier studies collected oysters from locations in the Chesapeake Bay, a rural estuary in Georgia, and the east coast of Florida. Wotton et al. [16] reviewed global trends of MPs in oysters to find an average biomass of MPs in oysters of 1.41 MPs/g ww, with wild-caught oysters having more MPs than those grown in aquaculture facilities.

Oysters are known for their sorting and preferential digestion of particles based on physical and chemical factors [64]. These selective feeders use their modified gills to sort plankton for ingestion and release the remainder in small pellets ("pseudofeces") to the sediment. Given oysters sort particles based on size and surface carbohydrates, they may have the capacity to sort by other material properties, including plastic type. Recent studies focusing on MPs in oysters (e.g., [8,65,66]) find that there are fewer MPs in oysters relative to the surrounding water, and those present do not directly reflect what is present in the surrounding environment. Collectively, studies are now showing that oysters ingest a subset of all MP particle types and/or selectively retain certain types of plastic. For example, in a detailed study by Ward et al. [67], a clear trend in the greater retention (or ingestion) of MP particles < 100 µm by eastern oysters was shown for microfibers (51% accumulation for 75 µm MPs) and microspheres (76% accumulation for 19 µm MPs). Future efforts in feeding experiments should consider exposing oysters to MPs coated in humic and other organic materials which will more likely reflect MPs in the natural environment, that is, rather than pure or nascent MPs which are typically purchased for this kind of work. Fabra et al. [68] have shown biofilm (produced by *Escherichio coli*)-coated microspheres (45 µm) to be ingested at a rate 3x higher than uncoated or control microspheres by European native oysters (*Ostrea edulis*) [68]. It remains to be determined whether larger particle sizes are made more palatable by being complexed with other

biofilms, such as exopolymeric substances produced by marine algae or other dissolved organic materials that are representative of the dynamics of coastal/marine environments.

4.1. Dominant MPs in the Bay and Oysters

Sources of plastics and the processes of their degradation differ by waterbody. This in turn affects plastic particle size and composition, which may affect the specific retention of particles in oyster tissues. Our results demonstrated that surface waters carried primarily fibers, similar to findings in other estuaries and coastal systems [69,70], including those in other parts of the Gulf of Mexico [63] and globally [4]. Synthetic clothing (polyester and acrylic) is recognized to be a significant source of MP pollution, with domestic washing machines releasing as much as 1900 fibers per garment in a single wash cycle [69]. MPs classed as fibers in this study were likely made of polyester and acrylic polymers, as reported by Napper and Thompson [71], which is also supported by our chemical analysis (see below). Fibers, fragments, foams, and films were found in both the Indian River Lagoon (FL, USA) water and *C. virginica* [63], with fibers being the dominant form as observed in the current study (Table 3). Comparing MPs in water samples to those in oysters, it appears that oysters in Galveston Bay had elevated counts of fibers, an order of magnitude higher than the other plastics measured (films, fragments, other). While Oakley et al. [35] also found fibers to dominate water grab samples in Galveston Bay, their sampling protocol generally reported higher densities of fragments. On the other hand, Walters et al. [63] reported that fibers compose 95.6% and 95.0% of MPs in water and oysters, respectively, in their study site in Florida. They found that colors were variable, but black fibers were the most common. In Galveston Bay, we found clear fibers were the most common color present, three-fold more common than black, four-fold more common than blue, and significantly more common than all other colors. In Brisbane (Australia), fibers were found to be the dominant material, particularly blue fibers [72].

In terms of chemical composition, ATR-FTIR measurements of surface waters found polyamide and polypropylene frequently, while ethylene propylene and polyethylene terephthalate were more commonly found in the lower parts of the bay. Py-GCMS/MS revealed four main types of plastics in Galveston Bay oysters regardless of location: polypropylene, nylon 66, polyethylene, and styrene butadiene. All these plastics are commonly found in global waterways. In Florida, Walters et al. [63] reported that polyethylene terephthalate was the most abundant polymer in lagoon water and oysters, composing 50%, and 56% of MPs, respectively. In the current study, polyethylene terephthalate was not detected in oysters. Instead, polypropylene and nylon66 were much more prevalent in oysters, while polyethylene and styrene butadiene (rubber) each made up ~20% of the MP load (Figure 7). The NMP composition in water samples from other Texas estuaries (Copano Bay, Aransas Bay, Corpus Christi Bay, Nueces Bay, Upper Laguna Madre, and Baffin Bay) measured using Py-GCMS/MS reported concentrations of MPs up to 1.6 µg/L [73], with the dominant forms being similar but not identical to those in the current study. Cisco et al. [73] found polyethylene, polypropylene and polyethylene terephthalate were the major components, with styrene-butadiene (rubber), nylon-66, and polymethyl methacrylate as the minor components in their water samples. Mortuza et al. [10] examined NMPs in oysters from Matagorda Bay (Texas) using Py-GCMS/MS; they found only three types of plastics: polypropylene, nylon 66, and polyethylene ($\sum$NMP = 8925 $\pm$2590 ug/g dw). By comparison, oysters in Galveston Bay had a higher (mean = 11,988 µg $\sum$NMP/g dw) and a broader range (172 to 67,783 µg $\sum$NMP/g dw) of NMP concentrations. In addition, styrene butadiene was present in Galveston Bay oysters but not in those in Matagorda Bay. Currently, it is not clear what factors are driving all the differences in NMPs along the Texas coast. Future studies might consider a coast-wide assessment to examine this question. The

values of NMPs from Matagorda and Galveston Bay oysters are in the same order of magnitude as those reported by Ribeiro et al. [72] for *Saccostrea glomerata* (rock oysters) deployed in the Brisbane River in Australia (Σ38,800 μg/g ww). If one assumes the ratio of wet to dry weight is 7.35 based on freeze-drying [74], then this is equivalent to 285 mg/g ww, at least an order of magnitude higher. In an earlier study, Ribeiro et al. [75] reported NMPs in *Crassostrea gigas* (oysters) to be Σ100 μg g^{-1} ww or Σ735 μg g^{-1} dw collected from the same location. In order to understand these discrepancies, several factors need to be considered. In the latter study, Ribeiro et al. [72] identified the 1–22 μm fraction to contain the highest total plastic mass concentration (79%), while the <1 μm and the >22 μm fractions had 20% and 1%, respectively, thereby revealing the importance of considering nanoplastics as part of the plastic body burden. This is only possible using Py-GCMS/MS, as ATR-FTIR and sorting/counting are limited to MPs. The other reasons for disparities may be related to the greater sensitivity of newer methods [10,73] and/or the higher accumulation rates of marine debris in some regions relative to others; this includes the Texas coast [31–33] which is home to Galveston Bay, located downstream of the nation's energy corridor and two major cities. Further, the environment in which oysters are grown influences their MP loads. We must not forget, fundamentally, the uniqueness of the environments in which the oysters are found. Ribeiro et al. [72] compared farmed versus deployed oysters, and the NMP loads were Σ115 ± 110 μg/g ww and Σ38,800 μg/g ww, respectively. Wotton et al. [16] and other reviews on MPs have revealed previously the importance of the environment in oyster MP studies.

Additionally, next steps should consider if ingestion of the NMPs is sufficient to inflict harm on oysters and/or if the adsorption of chemical pollutants to plastic particles creates another significant threat. Water samples contained polyethylene, the most widely used plastic in the world for products ranging from clear food wrap to automobile fuel tanks, and polypropylene, a high-volume commodity thermoplastic used in packaging, appliances and carpeting. Based on other studies, we know that polycyclic aromatic hydrocarbons (PAHs), polychlorinated biphenyls (PCBs) and other POPs are present in Galveston Bay, including its biota (oysters, fish) (e.g., [76]). It appears, though, based on the condition index, that N/MPs did not impact Galveston Bay oysters. However, going forward, examining if it is the degradation of N/MPs that is acting as a substrate upon which these POPs are more rapidly absorbed by the oysters remains a critical task for future studies.

4.2. Sources and Sinks of MPs

While rivers are thought to be the primary source of MPs in the ocean, studies examining the estuarine continuum between these sources and sinks are only now beginning to determine if these embayments act as a primary sink for settling material, and what the consequences may be to those that live their full life cycle in estuaries. Several recent studies have endeavored to model the abundance, transport, and biological interactions in estuaries including Galveston Bay, and the adjacent Gulf of Mexico [34]. Under normal hydrodynamic conditions, rivers and estuaries appear to retain MPs, releasing these into the ocean only under more extreme conditions, such as storms and floods [34,77]. In addition, MPs are frequently found many kilometers downstream of wastewater treatment plants, a known source of MPs, with concentrations exponentially higher in sediments than in the water column, indicating not only transport but also sequestration of MPs in rivers and estuaries where these facilities are typically located [70]. Walters et al. [63] found that tributaries were major MP sources while inlets were major MP sinks in the Indian River Lagoon (FL, USA), and there were more MPs in areas adjacent to urbanization. In the current study, we did not observe a spatial gradient in MPs in the water column or oysters; however, we do not preclude that under different environmental conditions (e.g., high

freshwater inflows) that such a gradient may be present. Indeed, Rakib et al. [78] found MP abundance was higher during the wet season than the dry season in the Karnaphuli River estuary, Bangladesh, with a gradient from upstream (lowest) to downstream (highest). Low pH and high temperature were found to be the key physiochemical factors determining MP distributions in this river. While they measured similar MP densities in oysters as those in the present study, Walters et al. [63] reported that the abundance of MPs in oysters was lower in the spring (and higher in the other seasons).

4.3. Impact on Oyster Health—Condition Index

Oyster CI provides a relative indication of health and is calculated by determining the amount of tissue within an oyster in relation to the available volume inside the shell [38,39]. Calculating oyster CI for each sampling location provided a first estimate of how spatial variation in MP loads within the estuary could be linked to spatial variations in oyster health. Many factors can influence the CI of oysters, such as seasonality, parasitism, water flow, nutrient availability, and spatial location [26,40,41,79]. Typical values for oyster CI range from 5 to 7 [38,40,41,79], with higher values indicating increased physiological health [38,40,80]. Despite there being fewer oysters in East Bay, the ones present were generally healthier than those in West Bay and Christmas Bay. These findings revealed that elevated levels of MPs found in oysters in Galveston Bay did not have an impact on oyster health. Differences in CI highlight spatial dynamics within the estuary consistent with previous studies on oyster population dynamics [26].

4.4. Impact on Human Health

Given MPs are being found in food sources, especially fish and shellfish, they are also present in humans [81–84]. A study estimated that up to 11,000 MP particles may be ingested annually by shellfish consumers in Europe and elsewhere around the world [57,85]. While most plastics are regarded as inert synthetic polymers, exposure may induce oxidative stress, inflammation, neurotoxicity, and reproductive toxicity, and/or change the structure of intestinal microflora in cells or biota [4]. Additional studies are needed to determine if this is because of the MPs themselves or if it may be because of the POPs adsorbed onto their surfaces [4,11]. Seafood consumers are becoming increasingly concerned about MPs, especially for species like oysters. It is therefore critical to provide baseline information for the public, including that which is specific to a given waterbody from which oysters are harvested. Given that ~70% of the oysters harvested in Texas annually come from Galveston Bay (https://www.nature.org/), scientists and citizens in this region are especially concerned. Perhaps more so given that a large portion of urban Texas (>10 million people; www.census.gov), from the Dallas-Fort Worth metroplex through Houston-Galveston, drains into Galveston Bay. In addition, in September 2019, aquaculture was legislatively allowed for the first time in the state, specifically for the farming of oysters. Hence, there will be an increase in the consumption of oysters from this estuary in the future.

Based on the present study on wild-caught oysters, Py-GCMS/MS revealed a very large range in concentrations of NMPs in oysters, from 28 to 10,925 µg NMP/g ww (or 172 to 67,783 µg NMP/g dw). This translates to an average daily intake of 0.85 ± 0.45 mg NMPs/Kg of body weight/day for adults, and a yearly intake estimated to be 310.0 ± 164.25 mg NMPs/Kg body weight/year. By implementing the ADI approach used in this study, the calculated average daily intake for humans consuming these oysters would be 1.2 mg NMPs/Kg body weight/day (or 434 mg NMPs/Kg body weight/year). In addition, Ribeiro et al. [75] calculated that ~0.7 mg of plastic will be consumed by humans eating just 10 oysters. An extensive review of the likely human toxicity of NMPs found

≤25 mg/Kg NMPs sufficient to be hepatotoxic when tested in vitro using human hepatic cell lines [83]. Of concern to human health is that levels ≤10,000 µg NMP/g have been reported in human tissues such as brain, kidney, and liver collected post-mortem [84]; measured using Py-GCMS/MS) and human blood [81,82]. These levels are comparable to those measured in oysters in our study and suggest that the exposure of humans to equivalent concentrations (as those measured in oysters) is likely. Given that potential (hepatic in vitro) toxicity is inducible at ~25 µg/g microplastics [83], there should be concern for an apparent 2000x bioaccumulation of NMPs in humans. Future management efforts need to consider screening oysters (wild-caught and farmed) for plastics before making them available for human consumption.

Collectively, this research suggests that oysters may be utilized as effective "passive" samplers of plastics in estuaries. Given their sessile nature and their selective feeding, comparing the MP content of similarly sized oysters (i.e., market-size) will allow scientists and managers to have a cumulative snapshot of plastic pollution in an estuary. This is important information given that oysters are a common food item in communities. This finding does not conflict with studies concluding that oysters are poor bioindicator species for MP pollution (e.g., [8,65,66]). In addition, this research as well as other publications consistently reveals that there continues to be poor alignment of polymers between oysters and their surrounding environment. Given that many chemical techniques are still emerging, the potential to use oysters as bioindicators requires continued improvement in these protocols, as well as an understanding that the traditional approach has challenges such as user experience and difficulties confirming materials are plastics versus other kinds of materials.

4.5. Methods Comparison

In the current study, three approaches were used to examine the nano- and microplastic content of oysters and surrounding surface waters. All these approaches require careful sampling in the field and laboratory, with recent reviews highlighting the main concerns for researchers [3,4,8]. Traditional (sorting, counting) approaches are widely used and focus on particles visible to the naked eye (>5 mm). While it is the least expensive approach in terms of equipment, it is the most labor intensive and requires many hours with trained personnel. ATR-FTIR can measure plastics that are several mm (present study) down to ~10 µm (micro-FTIR) and has the advantage of providing information on the composition of the MPs [51,56]. Common polymers identified include polyethylene, polypropylene, polyethylene terephthalate, and polyvinyl chloride. Recently, thermo-analytical techniques (e.g., Py-GCMS/MS) have been applied to NMP research [10,75,86]; this approach has higher analytical precision than other methods and no limitations on the size range or types of plastics in a sample. Of the three approaches, this method facilitates rapid screening of a wide range of samples (water, sediments, tissues). The Py-GCMS/MS method also provides the most comprehensive compositional analysis. The challenge with this latter approach is that the instruments are expensive and this limits the number of users which can consider this approach. In this paper we reported that these methods provided complementary information, but given their inherent differences, it was not unexpected that the findings were not identical. Ultimately, which approach to use is dependent on many variables, including affordability, time, and research goals. Given its broad application worldwide, the traditional approach has provided invaluable information to both the scientific community and public, raising awareness of MPs in a wide range of substrates. Both ATR-FTIR and Py-GCMS/MS have allowed us to not only measure MPs but also examine which kinds of plastics are most prevalent in oysters. Adding these approaches therefore provides

greater details which can benefit our interpretation of the consequences of plastics on biological systems.

4.6. Conclusions

This manuscript reports for the first time the body burdens of (N)MPs in oysters relative to their surrounding waters in Galveston Bay, located in the northwestern sector of the Gulf of Mexico. Unfortunately, some of the world's highest MP densities have been measured in this region [10,31–35,59]. Hence, there is an urgent need to understand their impact on local biological communities. Our key findings reveal there are 4–81 MPs/L (min–max) in surface waters, with no clear spatial patterns across the bay, as observed previously elsewhere [6,32,33,59,63–71]. Counts of MPs in oysters ranged from 0 to 200 MPs/oyster, consistent with global findings [16,46–48,66]. In surface waters and oysters, microfibers were found to be the most prominent type of MP, as reported elsewhere [5,6,16,87–92]. Further, ATR-FTIR and Py-GCMS/MS revealed the major types of plastics to be polypropylene > nylon 66 (N66) > and polyethylene [49], similarly to other studies in Texas [73] and globally [57,87–92]. Given these findings, we compared counts (converted to an equivalent w/v concentration) with analytical approaches to determine the body burdens of microfibers in oysters. This yielded microfiber body burdens of 9.1 + 1.2 µg/g ww or 56.4 + 7.3 µg/g dw. In contrast, Py-GCMS/MS revealed a larger range, from 28 to 10,925 µg ∑NMP/g ww oyster (or 172–67,783 µg ∑NMP/g dw). A key result from this study is the relatively close approximation of body burdens, i.e., within two orders of magnitude, from these disparate approaches. While our analysis of the condition index of oysters indicated no overtly adverse effects of MP body burdens [41,79,80], future studies should investigate sub-lethal biomarkers of toxicity and undertake a human exposure risk assessment [9,83–85]. In conclusion, the observed plastic densities in oysters and surface waters measured in this study contribute to the global understanding of the distribution and types of these materials and the growing body of literature raising awareness for international policy interventions to minimize ecological, social, and economic harm [85,87–92].

Author Contributions: Field collections, sample processing and data analysis were performed by M.C., M.H., L.J.J., M.K., A.M. and M.B.G. Microplastic counting, identification and data curation was conducted primarily by M.C. Funding was acquired by L.J.J., M.H., D.H., K.K. and A.Q. The first draft of the manuscript was written by A.Q., and all the authors, especially M.C., contributed edits to previous drafts. All authors have read and agreed to the published version of the manuscript.

Funding: This project has been funded in part by the United States Environmental Protection Agency (EPA) under grant number 00655007 to the Texas Commission on Environmental Quality (TCEQ) and flow-on funding to the Galveston Bay Estuary Program (Contract No. 582-21-10080) to Jurgens, Hanke and Quigg. The contents of this document do not necessarily reflect the views and policies of the EPA. The Py-GCMS/MS research presented in this manuscript was supported in part by the Matagorda Bay Mitigation Trust (Grant# 013) to Hala, Quigg, and Kaiser.

Data Availability Statement: Data has been submitted to the Texas Digital Library and is publicly available: https://doi.org/10.18738/T8/ML0CHU.

Acknowledgments: We thank the Texas Parks and Wildlife Department (TPWD) for collecting oysters used in the chemical analysis and J. Conkle (TAMUCC) for sharing his expertise in microplastics sample processing. We thank the many students and volunteers from the Jurgens lab for helping with sorting, identifying, and counting MPs, especially Chris Oxley, Karolina Parodi, and Emily Hubbard. We also thank the editor and reviewers, who provided excellent feedback which improved the final version of this paper.

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

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